

Logistic Regression Analysis of U.S. Income Levels: How Occupational Field Modifies the Income Effect of Attaining Advanced Education

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I. INTRODUCTION

Understanding of socio-economic factors that influence income levels and wealth disparity is fundamental to research in sociology, economics, and public policy. Income levels are indicators of economic and societal well-being.

U.S. Census data represents the survey used to scope the economic statuses of individuals across demographics nationwide, to inform local, state, and federal government decisions regarding public employment and finance [1]. Leveraging this data enables one to identify the primary drivers of income levels in the American workforce.

This study aims to assess the relative importance of education levels, types of occupations, and the number of hours worked per week on the probability of having a high income, as defined by earning over \$50,000 annually, using a predictive Logistic Regression framework. Our primary goal is to quantify which factors are most influential, testing the hypothesis that education level will be the most critical predictor, followed by type of occupation, then hours worked per week last. The results of this study are expected to provide empirical evidence of the strongest drivers of economic success, influencing prioritized efforts in educational and professional development.

II. METHODS & MATERIALS

A. Data Set

The dataset called 'Adult Census Income', sourced from U.S. Census Bureau data, was accessed via a public repository [2]. The raw dataset contains 32,561 observations and 15 variables.

This analysis used a subset of the original dataset. The outcome variable of focus was 'income'. The dataset was pre-cleaned for this variable, already having transformed the income values per individual into one of the two categories: >50K or ≤50K. I created a second column to transform this into a binary variable (HighIncome = 1, LowIncome=0) for later data processing.

The predictor variables consisted of two categorical and one continuous: *education*,

occupation, and *hours.per.week*. The *education* categorical values were categorized for clearer visualization and understanding, grouping similar categories (ie, all categories that represent ages earlier than obtaining a high school diploma were converted to "No High School Diploma", and "Advanced Degree" represents the values *Masters*, *Prof-school*, and *Doctorate*). I handled missing values, coded as ? in the raw data, by excluding rows containing missing values in the outcome variable or any of these 3 predictor variables. This resulted in a final sample size of 30,162 observations and a focus on 4 variables.

Table 1: Summary of Income and Predictors

Summarizes the studied variables. Binary outcome variable *income* prevalence is displayed along with the categorical values of the *education* and *occupation* variables. Summary statistics presented for *hours.per.week*.

Variable	Value	Count	Percentage
Outcome variable: income	LowIncome (0)	23068	75.09603
	HighIncome (1)	7650	24.90397
Predictor variables: education_group	No High School Diploma	3821	12.438961
	High School Graduate	9968	32.450029
	Some College/Associate	9116	29.676411
	Bachelor's Degree	5182	16.869588
	Advanced Degree	2631	8.565011
occupation (simplified)	Prof-specialty	4140	13.47743994
	Craft-repair	4099	13.34396771
	Exec-managerial	4066	13.23653884
	Adm-clerical	3770	12.27293444
	Sales	3650	11.88228400
	Other occupations	10993	35.78683507
hours.per.week	Min. 1.00 1st Qu. 40.00 Median 40.00 Mean 40.95 3rd Qu. 45.00 Max. 99.00		

B. Study Population

The study population consisted of 30,162 individuals representative of the U.S. adult workforce based on U.S. Census Data. The used data included all individuals recorded in the provided census data, but excluding approximately

2,300 or 7% of the original observations based on the missing-value exclusion criteria. The prevalence of high-income status (according to the >\$50,000 metric) was 24.9%. Table 1 provides the descriptive statistics for the 3 predictor variables. Note that *occupation* was simplified to the 5 most reported categories for readability of the table, but that the 14 categories represented in the dataset were used for modeling.

C. Statistical Methods

All analyses were conducted using R software version 4.2.3 within RStudio. Key packages for analysis were *tidyverse*, *car*, and *emmeans*.

Logistic Regression Model:

Using a generalized linear model (GLM) with binomial and a logit link function specified, I modeled the probability of an individual having a high income through a multiple Logistic Regression model. This model used the three predictors (education, occupation type, and hours.per.week) and an interaction term of education and occupation (the two categorical variables). The interaction term was used to study whether the association between income and education varied across different career fields.

Type III ANOVA Test:

To test the relative importance hypothesis via analysis of unique contributions per predictor variable, I used a Type III Wald χ^2 Test. This was performed on the fitted logistic regression model mentioned above using the `Anova()` function from the *car* package. This quantifies the effect of unique predictors, controlling for the presence of other variables, which establishes a ranking of the importance of predictor variables.

III. RESULTS

Model Fit and Relative Importance of Predictors

The full multiple Logistic Regression model was highly significant (LR $\chi^2 = 6329.0$, $p < 0.001$), confirming that the set of predictors successfully distinguishes high and low-income individuals. The predictors were ranked by their Likelihood Ratio Chi-Squared (using `Anova`). The following conclusions were drawn with a small p-value of 0.001:

1. The number of hours worked per week was found to be the single most influential predictor ($\chi^2 = 727.50$).

2. Occupation was found to be the second most critical ($\chi^2 = 115.32$).
3. Education level was found to be the third most important ($\chi^2 = 80.83$).

The ranking of the Type III Wald χ^2 test *rejects* the hypothesis that education level is the primary predictor of high-income status. Importantly, the education/occupation interaction term with χ^2 value of 103.60 ($p < 0.001$) implies that the income effect of education also varies across different occupations/career fields.

Table 2: Odds Ratios

This table represents the magnitude of effect (ie, the largest Odds Ratio represented, 13.265 for Advanced Degree, indicates that there is the largest effect in comparison to the reference variable).

Predictor Variable	Odds Ratio	95% Confidence Interval	p-value
Hours per week (per hour increase)	1.036	(1.034, 1.039)	0.001
Education (Reference: 'No High School Diploma')			
High School Graduate	3.542	(1.539, 8.152)	0.003
Some College/Associate	3.584	(1.561, 8.231)	0.003
Bachelor's Degree	7.955	(3.423, 18.485)	0.001
Advanced Degree	13.265	(5.196, 33.864)	0.001
Occupation (Reference: 'Adm-clerical')			
Armed-Forces	0.000	(0.000, inf)	0.990
Craft-repair	2.542	(1.080, 5.986)	0.033
Exec-managerial	6.663	(2.617, 16.960)	0.001
Farming-fishing	0.789	(0.280, 2.229)	0.655
Handlers-cleaners	1.032	(0.389, 2.741)	0.950
Machine-op-inspct	1.198	(0.478, 3.003)	0.700
Other-service	0.391	(0.142, 1.076)	0.069
Priv-house-serv	0.000	(0.000, 6.304e+82)	0.914
Prof-specialty	4.130	(1.306, 13.060)	0.016
Sales	2.373	(0.950, 5.925)	0.064
Tech-support	2.624	(0.465, 14.800)	0.274
Transport-moving	3.044	(1.263, 7.337)	0.013

Odds Ratios

The Odds Ratios of all predictors are seen in Table 2. These values indicate that each one-hour increase in a work week increases the likelihood of having a high income by 3.6% (1.036, $p < 0.001$). Over the span of a median 40.00-hour week, this means the effect is $1.036^{40.00}$ or approximately 4.15.

The impact of education level on high income is visualized by the Estimated Marginal Means plot in Figure 1, which displays the predicted probability rising from 1.3% for the reference group ('No High School Diploma') to 21.5% for the 'Advanced Degree' group.

Estimated Marginal Means were calculated from the logistic regression model using the *emmeans* package. This estimates the

model-predicted probability of high income for each level of education, holding other variables constant at their mean or reference level.

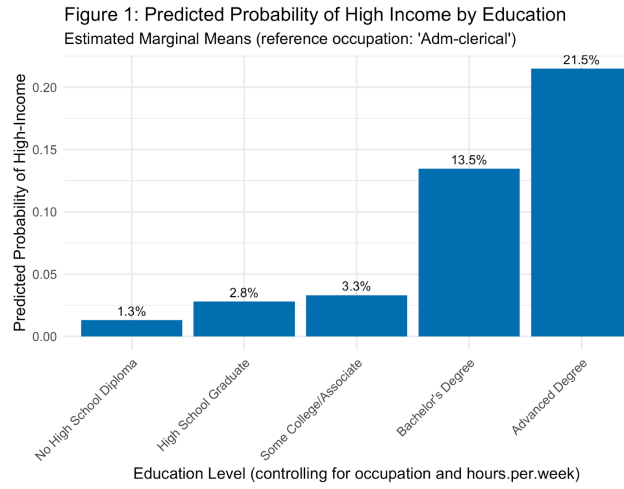


Figure 1: The figure displays the Estimated Marginal Means, representing the predicted probabilities for each education group. These probabilities are adjusted for the effects of occupation (held at the reference level 'Adm-clerical') and hours.per.week (held at the sample mean).

There was variation in the odds ratios for occupations, highlighted in Figure 2. The highest likelihoods were found in 'Exec-managerial' and 'Prof-specialty', and the lowest stable likelihoods were in the 'Other-service' category. 'Armed-forces' and 'Priv-house-serv' have Odds Ratios listed as 0.00, due to a low number of high-income earners within these two categories in the analytical sample (resulting in quasi-complete separation).

Figure 2: Odds Ratios by Occupation

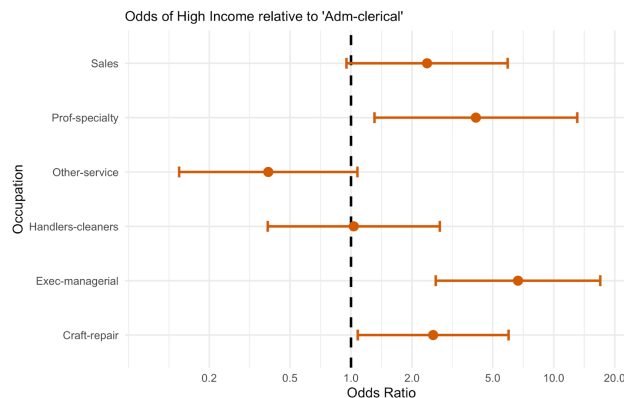


Figure 2: Display of adjusted odds of having high income for a selection of occupational groups with horizontal bars representing the 95% Wald Confidence Intervals.

IV. CONCLUSIONS

The findings of this study *reject* the hypothesis that education level is the primary predictor of high income. The analysis established that the hours.per.week variable, or the amount of labor time per week, gives the greatest unique contribution to predicting high income, followed closely by the type of occupation. However, there is a strong significance of the Education * Occupation interaction term which creates nuance. While the hours worked per week variable is the primary driver of income variance, the economic returns for investment in education (ie, obtaining an advanced degree) are not guaranteed across occupations.

V. DISCUSSION

While the amount of time worked is the major contributor to income variance, the strong significance of the Education*Occupation interaction term introduces something to consider when comparing these conclusions to similar studies. The economic return on educational investment is conditional on the specific career field, as exemplified by the large Odds Ratios for an Advanced Degree. This does not imply that having an advanced degree inherently predicts high income. The interaction term instead indicates that the economic return for attaining the advanced degree is not uniform but conditional on the specific career field (*occupation* variable). For example, an advanced degree in conjunction with an occupation under "Exec-managerial" is a strong predictor for High Income. Conversely, the "Other-service" occupation, which represents jobs in the service industry, displays an environment where higher education levels are largely negated and do not imply a high income.

Thus, simply obtaining any level of degree is insufficient to predict high income; the financial benefit derived from a higher level of education is significantly amplified or diminished by the occupational context in which the education is applied.

The findings of this study have some similarity with those of another study by the World Economic Forum, asking, "*Do top earners work more hours?*" [3]. Their conclusion, "America's top 10% of earners work an average of 4.4 hours more each week than those in the bottom 10%, according to a US Census Bureau survey, [3]" is based upon the same type of

survey data and additionally portrays that a greater number of hours can be indicative of a high income status.

ACKNOWLEDGEMENTS

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REFERENCES

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<https://www.census.gov/about/what.html#:~:text=The%20U.S.%20census%20counts%20every,%2C%20state%2C%20and%20local%20levels.>
2. Lovish Bansal, "Adult Census Income," *Kaggle*, 2023.
<https://www.kaggle.com/datasets/lovishbansal123/adult-census-income/data>
3. Urban Transformation, "Do top earners work more hours? It depends which country they live in," *World Economic Forum*, 2022.
<https://www.weforum.org/stories/2022/09/worki ng-hours-america-income-economy/>

APPENDIX

I. Source Data File

This dataset was a cleaned version published by Lovish Bansal on Kaggle that was representative of U.S. Census data collected by the U.S. Census Bureau via survey.

II. Data set

The dataset generated from my R code has been uploaded to BruinLearn as *Skye_Pickett_dataset_income.csv*.

The following data cleaning was conducted to produce the uploaded dataset:

```
df_clean <- data_raw %>%
  drop_na(income, education, occupation,
hours.per.week) %>%
  select(income, education, occupation,
hours.per.week) %>%
  mutate(
    income = factor(income, levels =
c("<=50K", ">50K"), labels =
c("LowIncome", "HighIncome")),
```

```
income_binary = as.integer(income ==
"HighIncome"),
education = as.factor(education),
occupation = as.factor(occupation),
education_group = case_when(
  education %in% c("Preschool",
"1st-4th", "5th-6th", "7th-8th", "9th",
"10th", "11th", "12th") ~ "No High School
Diploma",
  education == "HS-grad" ~ "High
School Graduate",
  education %in% c("Some-college",
"Assoc-voc", "Assoc-acdm") ~ "Some
College/Associate",
  education == "Bachelors" ~
"Bachelor's Degree",
  education %in% c("Masters",
"Prof-school", "Doctorate") ~ "Advanced
Degree",
  TRUE ~ as.character(education)
),
education_group =
factor(education_group, levels = c("No
High School Diploma",
"High School Graduate",
"Some College/Associate",
"Bachelor's Degree",
"Advanced Degree"))
)
```

III. R Code

Table 1:

```
## Summary statistics

summary_hours <-
summary(df_clean$hours.per.week)
sd_hours <- sd(df_clean$hours.per.week)

# frequency tables

table_education <-
table(df_clean$education_group)
```

```
prop_education <- print(table2_data_clean)
prop.table(table_education) * 100

table_occupation <-
table(df_clean$occupation)
prop_occupation <-
prop.table(table_occupation) * 100

income_prevalence <-
table(df_clean$income)
prop_income_prevalence <-
prop.table(income_prevalence) * 100

Table 2:
# Odds Ratios Table
model_summary <-
summary(model_logreg)$coefficients

zscore_95 <- 1.96

# +/- Z * se

log_odds_estimate <- model_summary[,
"Estimate"]
log_odds_se <- model_summary[, "Std.
Error"]

log_odds_lower_ci <- log_odds_estimate -
(zscore_95 * log_odds_se)
log_odds_upper_ci <- log_odds_estimate +
(zscore_95 * log_odds_se)

table2_data <- data.frame(
  Coefficient=rownames(model_summary),
  Estimate=log_odds_estimate,
  P_Value=
summary(model_logreg)$coefficients[,
"Pr(>|z|)"],
  Odds_Ratio = exp(log_odds_estimate),
  Lower_CI = exp(log_odds_lower_ci),
  Upper_CI = exp(log_odds_upper_ci)
)

table2_data_clean <- table2_data %>%
  filter(Coefficient != "(Intercept)")
%>%
  mutate(across(where(is.numeric), ~
round(., 3)))
```

```
print(table2_data_clean)
```

Figure 1:

```
emm_education <- emmeans(model_logreg, ~
education_group, type = "response")
```

```
emm_education_df <-
as.data.frame(emm_education)
```

```
print(emm_education_df)
```

```
library(ggplot2)
```

```
library(dplyr)
## visualizations
```

```
emm_education_df <- emm_education_df %>%
  mutate(prob_percent = round(prob
*100,1))
```

```
figure1 <- ggplot(emm_education_df, aes(
x = education_group, y = prob)) +
  geom_bar(stat = "identity", fill =
"#0072B2") +
  geom_text(aes(label =
paste0(prob_percent, "%")), vjust = -0.5,
size=4) +
  labs(
    title = "Figure 1: Predicted
Probability of High Income by Education",
    subtitle = "Estimated Marginal Means
(reference occupation: 'Adm-clerical')",
    x = "Education Level (controlling for
occupation and hours.per.week)",
    y = "Predicted Probability of
High-Income"
  ) +
  theme_minimal(base_size = 14) +
  theme(axis.text.x =
element_text(angle=45, hjust = 1))
```

```
figure1
```

Figure 2:

```
figure2_data <- table2_data %>%
```

```
filter(
  Coefficient %in% c(
    "occupationExec-managerial",
    "occupationProf-specialty",
    "occupationCraft-repair",
    "occupationSales",
    "occupationOther-service",
    "occupationHandlers-cleaners"
  )
) %>%
mutate(
  Occupational_Group =
gsub("occupation", "", Coefficient)
)

ref_line <- 1

figure2 <- ggplot(figure2_data,
aes(y=Occupational_Group, x = Odds_Ratio,
xmin= Lower_CI, xmax = Upper_CI)) +
geom_vline(xintercept = ref_line,
linetype = 'dashed', color = 'black',
size = 1.2) +
geom_point(size=4, color = "#D55E00") +
geom_errorbar(width=0.2, linewidth = 1.2,
color = "#D55E00") +
labs(
  title = "Figure 2: Odds Ratios by
Occupation using 95% Wald CI",
  subtitle = "Odds of High Income
relative to 'Adm-clerical'",
  x = "Odds Ratio",
  y = "Occupation"
) +
  scale_x_log10(breaks = c(0.2, 0.5, 1,
2, 5, 10, 20), limits = c(0.1, 20)) +
  theme_minimal(base_size = 14)

figure2
```